# Report R23 — the stripes are walls, the flicker is a dark qubit, and only rules 73 and 109 are doing many-body physics

Krylov sector analysis of quantum cellular automata

Tier 1 analysis

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**Abstract**

The trajectory pictures for $W_{28}$, $W_{29}$, $W_{70}$, $W_{71}$, $W_{157}$ and $W_{199}$ all end in a stripe pattern whose gaps appear to breathe slightly, which raises the fair suspicion that these six rules are trivial. They are — *asymptotically*, and now for a stated reason rather than an impression. Every terminal SCC of all six is a **subcube**: its states agree on a set of frozen sites and range freely over the rest, so $|A| = 2^{\#\mathrm{free}}$ (exhaustive, $N = 6 \ldots 14$). Together with R17's completeness this says exactly one thing — the attractor is $k$ independent unbiased coins and nothing else. The frozen part is a **wall tiling**: the attractor words are precisely the tilings of the chain by `10` and `f10` (rules 199, 71; mirrored for 157, 29) or by `10`, `1f0` plus a leading vacuum run (rules 28, 70), as set equality for $N = 6 \ldots 14$. That *derives* R17's two fitted constants: tiles of length 2 and 3 give the plastic number $\rho$, and the all-wide tiling gives $2^{\lfloor (N+1)/3 \rfloor}$. Every attractor of all eight rules is moreover an exact decoherence-free subspace — the jump probability on it is 0 identically — so the channel restricted to an attractor is a *unitary*. For the six that unitary is a product of single-qubit Hadamards on the free sites: each runs $H^t$ and oscillates with **period** 2. The stripes are the walls; the breathing gap is one dark qubit per wide tile; the half-chain entanglement is identically zero because the attractor subspace is a tensor product across every cut (membership-matrix rank 1). The period-2 signal is single-shot: averaging over trajectories drops it from $\sim 0.45$ to $\sim 0.05$, so it is invisible in an averaged plot. Rules 29 and 71 — and only they — have weak components holding several terminal SCCs ($n_{\mathrm{wcc}} = n_{\mathrm{rec}}$ exactly for the other six). The selection is a set of independent **fair coins resolved in one period**: absorption probabilities are uniform on $2^k$ attractors with $k \in \{1, 2\}$, never anything else. It is multistability with noise-induced selection, not a bifurcation. Finally $W_{73}$ and $W_{109}$ are a different animal and the answer to "are they more interesting" is yes: their attractors are *not* subcubes but the hard-core constrained spaces — no two adjacent 1s for $W_{73}$ ($\dim = F(N+2)$), no two adjacent 0s for $W_{109}$ ($\dim = F(N)$). The dissipation prepares an exponentially large decoherence-free subspace carrying a kinetically constrained unitary, and the state inside it is entangled.


## Contents

# 1 The question

Rules 28, 29, 70, 71, 157, 199 are R17's plastic class. In the trajectory simulations they all relax to a stripe pattern with what looks like a slow oscillation of the space between stripes. Three things were asked: whether that is really what happens, how it relates to the terminal-SCC digraph, and what precisely goes on in 29 and 71, whose weak components do not have unique terminal SCCs. A fourth question asked what 73 and 109 look like and whether they are more interesting. All four are answered below; the last one has the largest answer.

Everything here is at `obc0` with $V = H$, and every statement labelled *exhaustive* is a full enumeration of $2^N$, not a sample.

# 2 The stripes are walls

## 2.1 Every attractor is a subcube

Write an attractor $A$ as a word over $\{0, 1, \mathtt{f}\}$: site $i$ gets its common value if every state of $A$ agrees there, and $\mathtt{f}$ if it does not.

**Proposition 1.** *For all six rules and $N = 6, \ldots, 14$, every terminal SCC satisfies $|A| = 2^{\#f}$ — $A$ is the full subcube on its free sites.*

This is the missing half of R17. R17 showed each terminal SCC is a complete digraph $K_n$; "$K_n$ on a subcube" is precisely the statement that the free bits are redrawn independently and uniformly every period. The graph is the product of $k$ unbiased coins, which is why its second eigenvalue is exactly 0 and why $n$ is always a power of two.

### 2.2 The frozen part is a tiling, and it fixes both constants

A tile is a wall `1` together with the empty sites that belong to it; a wide tile carries one free site.

**Proposition 2.** *The set of attractor words is* exactly *the set of tilings of the chain by*

| | |
|---|---|
| $W_{199}$, $W_{71}$ | *`10`, `f10`* |
| $W_{157}$, $W_{29}$ | *the mirror image of those* |
| $W_{28}$, $W_{70}$ | *`10`, `1f0`, preceded by a run of vacuum tiles `0`* |

*with two edge conditions. The chain may end inside a tile, but the remnant must still contain its wall — a free site with no wall beside it is not dark, so `f10` may be cut to `f1` and never to `f`. And for $W_{28}$, $W_{70}$ the chain must end* at *a wall: a trailing empty site with nothing after it is refilled, so the final tile is always the truncated one, whereas the* 199 *family may end on a full tile. Set equality — no containment either way — for $N = 6, \ldots, 14$, and it is these two conditions that make it equality rather than containment.*

Two consequences drop out without a fit.

*The plastic number.* Tiles of length 2 and 3 give the counting recursion $a(N) = a(N-2) + a(N-3)$, whose growth rate is $\rho = 1.3247$, the root of $x^3 = x + 1$. The counts are $5, 7, 9, 12, 16, 21, 28, 37, 49$ for the 199 family and $11, 15, 20, 27, 36, 48, 64, 85, 113$ for the 28 family over $N = 6 \ldots 14$, matching R17's series term for term. The 28 family is larger by exactly the leading vacuum run, and it is the *absence* of an `E` that permits it: with an `E` at $(0, 0)$ the empty region refills, so 157, 199, 29, 71 have no vacuum tile. That is R17's "the children carrying an `E` keep a strict subset", now with the reason attached.

*The base* $2^{1/3}$. The largest attractor is the all-wide-tile word, so its dimension is $2^{\lfloor (N+1)/3 \rfloor}$ exactly, giving $b_{\text{att}} = 2^{1/3}$ as an identity rather than a fitted constant.

## 3 Every attractor is decoherence-free

**Proposition 3.** *For all eight rules, on every terminal SCC tested the jump probability is* 0 *to machine zero at every site of every period, the no-jump Kraus operator removes no norm ($\leq 3 \times 10^{-16}$), and the support never leaves the SCC. The channel restricted to a terminal SCC is therefore a* unitary*.*

This is the sharp form of R17's "the reset never fires on its own attractor". The dissipation is spent entirely on the transient; what it leaves behind is a decoherence-free subspace. The interesting question then becomes which unitary, and the six rules and the two rules answer it differently.

### 3.1 Which unitary: always the coherent parent, for all eight

The answer is the same for every one of the eight, and it is the parent rule. Each dissipative rule is its coherent parent with one identity replaced by a reset, and on the parent's own sectors that reset never fires.

**Proposition 4.** *For each of the eight, every non-trivial terminal SCC is* exactly *a Krylov sector of the coherent parent — set equality, not inclusion — and on it the child's one-cycle matrix equals the parent's to $\leq 2.2 \times 10^{-16}$, with no amplitude leaving the subspace. Verified on* every *non-trivial attractor at $N = 9$ and* 11 *—* 21 *and* 41 *of them for $W_{28}$ and $W_{70}$,* 11 *and* 20 *for $W_{157}$, $W_{199}$, $W_{29}$, $W_{71}$,* 24 *and* 55 *for $W_{73}$,* 27 *and* 68 *for $W_{109}$ — and, for* 73 *and* 109*, on the whole constrained space at $N = 4, \ldots, 11$. Counts independently reproduced by the Tier-1 session.*

This upgrades R17, which established the same statement at the level of the support graph — same vertices, same edges — to the level of the operator.

The four parents are the four single-`V` rules, and they differ only in which neighbour pattern gates the Hadamard:

| parent | word | fires when | dressed form |
|---|---|---|---|
| $W_{201}$ | `VIII` | both neighbours empty | $P^0HP^0$ |
| $W_{108}$ | `IIIV` | both neighbours occupied | $P^1HP^1$ |
| $W_{156}$ | `IIVI` | left occupied, right empty | $P^1HP^0$ |
| $W_{198}$ | `IVII` | left empty, right occupied | $P^0HP^1$ |

So all four are projector-dressed Hadamard Floquet models and the whole difference between the two families is whether the projector reads a *symmetric* pattern or a *domain wall.* Symmetric (00) or (11) is a blockade: the invariant space is the hard-core constrained space, of dimension $F(N+2)$, and the dynamics on it entangles. Asymmetric (10) or (01) is a domain-wall detector: the invariant spaces are the tiling subcubes, and the dynamics on them does not entangle at all. That single slot is the entire difference between §4 and §6.

**Proposition 5.** *On every terminal SCC of the six, the induced unitary is exactly the tensor product* $\bigotimes_{i\in\text{free}} H_i$ *of single-site Hadamards on the free sites — agreement* $1.1\times10^{-16}$ *at every one of the* 21/41 *and* 11/20 *non-trivial attractors (*$N=9$ / $N=11$*).*

Period 2 and zero entanglement are then both immediate: $H^2=\mathbb{1}$ site by site, and a tensor product of single-site gates on a product subspace cannot correlate anything. The six are not merely *like* decoupled qubits; on their attractors they are decoupled qubits, exactly.

# 4 The flicker: one dark qubit per wide tile

For the six, the free sites are isolated — each sits between a wall and an empty site — so the surviving unitary is a product of single-qubit Hadamards, one per wide tile. Each free qubit runs $H^t$, and since $H^2=\mathbb{1}$ it cycles $|0\rangle\to|+\rangle\to|0\rangle$: the occupation at that site alternates between a definite value and 1/2 with **period** 2, forever, exactly. Measured deep in the attractor, $\max_t|n(t+2)-n(t)|=0$ to machine precision at every $N$ tested and in both brickwork conventions (engine even-first, `pyqca` odd-first).

So the trajectory picture decomposes cleanly:

| what you see | what it is |
|---|---|
| bright stripes, fixed | the walls of the tiling |
| dark gaps of width 1 or 2 | `10` and `f10` tiles |
| the gap that breathes with period 2 | the dark qubit of a wide tile |
| which stripe pattern you get | which tiling the noise selected |

## 4.1 The entanglement is exactly zero, not just small

Being a subcube, the attractor subspace is a tensor product of single-site spaces, hence a tensor product across *every* cut. Checked directly at the middle cut: the membership matrix of left-half against right-half configurations has rank 1 for all 41/77 (rule 28) and 20/36 (rule 199) non-trivial attractors at $N=11,13$, and likewise for 70, 157, 29, 71. Hence *every* state in the attractor is a product state, and the measured half-chain entropy of the trajectory is 0 identically — not statistically zero, deterministically zero, at every $N$ and on every trajectory.

## 4.2 What survives averaging, and what does not

Two different things happen to the two features, and they should not be lumped together.

The *oscillation* is destroyed. The period-2 amplitude of a single trajectory is 0.42–0.50, essentially its maximum of 1/2; averaged over 200–400 trajectories it collapses to 0.03–0.07. Different trajectories land on different tilings and reach them at different times, so the phases add incoherently.

The *stripes* are damped but not erased. A single trajectory has full contrast, $\max_i n_i - \min_i n_i = 1.000$: the frozen sites sit at exactly 0 or 1. Averaged over 300 trajectories at $N = 15$ the contrast falls to 0.48 ($W_{28}$) and 0.31 ($W_{199}$), and what survives is not any one tiling but a staggered residue — a period-2 modulation in *space* about $n \approx 0.5$ with the boundary sites strongly pinned (0.81 at the right edge for $W_{28}$). That residue is a consequence of the grammar: walls are separated by 2 or 3 sites, so across the tiling ensemble they still favour one sublattice, and the open boundary fixes the first wall. For 73 and 109 the same average is much flatter, contrast 0.12 and 0.16.

**Prediction for the notebook:** the cells plotting one trajectory should show crisp full-contrast stripes with a visible flicker; the cell taking `np.mean` should show a much weaker staggered pattern about $n \approx 1/2$ with bright edges, and no time dependence at all. That is a check on this report, not a result of it.

# 5 Rules 29 and 71: several enclosures in one block

## 5.1 Only these two split

Counting weak components against terminal SCCs at $N = 9, \ldots, 13$:

| rule | 28, 70 | 157, 199 | 73, 109 | 29 | 71 |
|---|---|---|---|---|---|
| $n_{\text{wcc}}$ vs $n_{\text{rec}}$ | equal | equal | equal | 7 vs 16 ($N$=10) | 12 vs 16 ($N$=10) |

Six of the eight have an exact bijection: one attractor per weak component. The gap is a statement about *which rules* can split, not about every size: it is absent below $N = 6$ for $W_{29}$ (which has it at $N = 4$, loses it at $N = 5$, and keeps it from $N = 6$ on) and below $N = 7$ for $W_{71}$. Over $N = 7, \ldots, 14$ both show it at every size. Cross-checked against the Tier-1 session's independent enumeration at $N = 4, 5$, which agrees row for row. In the R22 dictionary a weak component is a strong-symmetry block and a terminal SCC is an enclosure, so those six have a *unique* steady state inside every block. Rules 29 and 71 do not: a block can carry up to six enclosures, and the asymptotic state depends on the trajectory.

## 5.2 What happens is a fair coin, resolved in one period

**Proposition 6.** *For $W_{29}$ and $W_{71}$ at $N = 8, \ldots, 12$: every state whose fate is undecided has* all *of its successors already committed, so the branch resolves in exactly* ***one period****; and the absorption distribution is uniform on $2^k$ attractors with $k \in \{1, 2\}$ — probability vectors are* $(\frac{1}{2}, \frac{1}{2})$ *(560 states at $N = 12$) or* $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$ *(8 states), and nothing else ever occurs.*

The shared-route states are 1.6%–14% of the space $(4, 24, 32, 128, 192$ for $W_{71}$ at $N = 8 \ldots 12)$ and are all transient.

## 5.3 What the coin chooses, and why it needs both resets

Take $W_{29}$ at $N = 9$ and the branch point `110001000` (site 0 leftmost). In one period it reaches eight states with probability 1/8 each, four in each of two attractors. Tracing the layers: on the

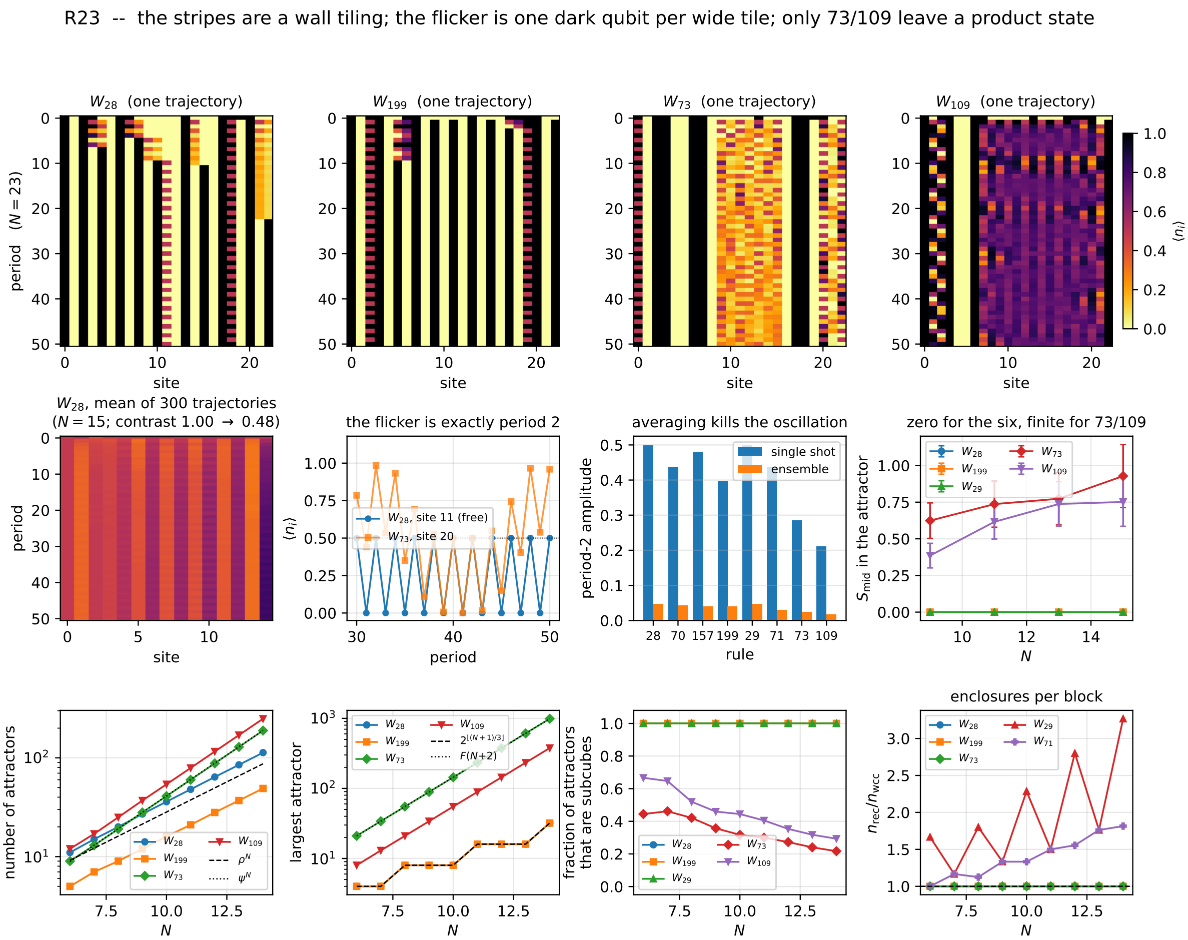


Figure 1: **Top row:** one quantum trajectory each at $N = 23$, the size used in `pyqca/mixed/trajectories.ipynb` (odd layer first, fixed-zero boundary). $W_{28}$ and $W_{199}$ lock into a wall tiling; $W_{73}$ and $W_{109}$ never do. **Middle row, left to right:** the same rule averaged over 300 trajectories, where the stripes wash out because each run picks a different tiling; the occupation of one free site, alternating with period 2 against a site of $W_{73}$ that does not; the period-2 amplitude single-shot against ensemble-averaged; and the half-chain entanglement inside the attractor, which is identically zero for the six and finite for 73 and 109. **Bottom row:** the number of attractors against $N$ with $\rho^N$ and $\psi^N$ for reference; the largest attractor against $2^{\lfloor (N+1)/3 \rfloor}$ and $F(N+2)$; the fraction of attractors that are subcubes — exactly 1 for the six, well below it for 73 and 109; and the number of enclosures per strong-symmetry block, which is 1 for every rule except 29 and 71.

even layer site 2 sees $(m, n) = (1, 0)$ and gets a Hadamard — that is the coin — and site 8 sees $(0, 0)$ and is excited by the `E`. On the odd layer the outcome at site 2 routes everything else:

- site 2 measures 1: site 3 then sees $(1, 0)$ and is Hadamarded into the free site of a wide tile, while site 1 sees $(1, 1)$ and is *reset by the* `D`. The wall lands at site 2, giving the tiling $(2, 3, 3, 1)$.
- site 2 measures 0: site 3 sees $(0, 0)$ and is *excited by the* `E` into a new wall, while site 1 is Hadamarded into the free site. The wall lands at site 3, giving the tiling $(3, 2, 3, 1)$.

So the coin exchanges a short tile with a wide one, and it needs *both* resets: the `E` writes the new wall, the `D` erases the old one. That is exactly what $29 =$ `EIVD` and $71 =$ `EVID` have and what $28 =$ `IIVD`, $70 =$ `IVID`, $157 =$ `EIVI`, $199 =$ `EVII` do not — which is why only these two split.

### 5.4 What survives as a quantum number

The block is not label-free. Across all 20 multi-attractor blocks of $W_{29}$ and all 16 of $W_{71}$ at $N = 8, \dots, 12$, the **number of walls is constant** — that is the strong symmetry the extra reset leaves intact. Their *arrangement* is not: the coin permutes tiles. For $W_{71}$ the multiset of tile lengths is conserved as well (16/16), so the branch is a pure rearrangement, e.g. a block whose three attractors are the tilings $(3, 2, 2, 1)$, $(2, 3, 2, 1)$, $(2, 2, 3, 1)$. For $W_{29}$ the multiset survives in only 8/20 blocks, because its truncated boundary tile can trade a unit — one block at $N = 10$ holds $(2, 2, 2, 2, 2)$ alongside $(2, 2, 3, 2, 1)$, same five walls, different widths. The two rules are a reflection pair and `obc0` with a fixed layer order is not reflection-covariant for dissipative rules, so an asymmetry at the right edge is expected rather than surprising.

### 5.5 Is it a bifurcation? No

A bifurcation is a qualitative change of the attractor set as a parameter is varied. Nothing is varied here: there is no coupling to tune, the branching ratio is exactly 1/2 at every $N$, and the attractor set is the same tiling family as for 157 and 199 — rules 29 and 71 have *identical* attractor words to 157 and 199 respectively; the extra `D` changes only which basins are joined. The correct name is multistability with noise-induced selection: several enclosures inside one strong-symmetry block, chosen by the measurement record.

Nor is the branch coherent. Starting the unmonitored trajectory *at* the branch point, 40 runs all end supported in a single terminal SCC with $S_{\text{mid}} = 0$ (19 one way, 21 the other) — the `D` acts as a which-path detector, so the ensemble is a classical $\frac{1}{2}/\frac{1}{2}$ mixture of two stripe patterns, not a superposition of them.

### 5.6 Coherence between attractors survives too

§3 was stated one attractor at a time, which understates it. The no-jump Kraus operator is a single global operator serving every enclosure at once, so by linearity it annihilates the jump branch on the whole recurrent space, not merely on each piece. Checked directly: prepare $(|a\rangle + |b\rangle)/\sqrt{2}$ with $a$ and $b$ drawn from *different* terminal SCCs, and over 20 periods the jump probability stays 0 exactly, the support stays in $A \cup B$, and the weight stays at 0.5000/0.5000 — for all six rules tested. A superposition of two different stripe patterns is preserved; the machine's own resets never decohere it.

Assembling the induced map on the whole recurrent space makes the period-2 result algebraic rather than numerical. For the six, that map has *exactly two* distinct eigenvalues and satisfies $U^2 = \mathbb{1}$ to $2.2 \times 10^{-16}$ — e.g. $D = 9$ with multiplicities $(6, 3)$ for $W_{28}$ at $N = 4$, $D = 11$ with $(6, 5)$ for $W_{199}$ at $N = 6$. Period 2 is then forced, not measured. For $W_{73}$ at $N = 5$ the same map has 12 distinct eigenvalues with multiplicities $(8,4,2,2,1^8)$ — stable under clustering tolerances from $10^{-10}$ to $10^{-2}$ — and $\|U^2-\mathbb{1}\|=0.5$, so no period, as §6.4 found.

This also settles which of these rules could hold a code. It cannot be any of them: a subspace spanned by computational basis states cannot detect a single-site $Z$, because the $W^\dagger Z_i W$ are diagonal with $\pm1$ entries given by the $i$-th bit and distinct codewords are distinct bit strings, so every joint eigenspace is one-dimensional. All eight attractors are basis-spanned — subcubes for the six, constrained subsets for 73 and 109 — so all have distance 1 and no subcode helps. The honest description is a noiseless subsystem against *this machine's own resets* and nothing else. (Argument due to the Tier-1 session, R24.)

# 6 Rules 73 and 109 are doing something else

## 6.1 The attractors are constrained spaces, not subcubes

Only 13/55 and 21/68 of the non-trivial attractors of $W_{73}$ and $W_{109}$ at $N=11$ are subcubes, against 100% for the six. The largest one is the diagnostic:

| rule | largest attractor | dimension |
|---|---|---|
| $W_{73}$ | all bitstrings with **no two adjacent** 1s | $F(N+2)$ |
| $W_{109}$ | all bitstrings with **no two adjacent** 0s | $F(N)$ |

Verified for $N=7,\ldots,13$: $W_{73}$ gives $34,55,89,144,233,377,610$ and $W_{109}$ gives $13,21,34,55,89,144,233$. This is the hard-core / blockade constrained space, and it is where $b_{\mathrm{att}}=\varphi$ comes from. The number of attractors grows at the supergolden ratio $\psi=1.4656$ $(13,19,28,41,60,88,129$ for $W_{73})$.

Combined with §3, the physical statement is: **the dissipation prepares an exponentially large decoherence-free subspace which is a kinetically constrained Hilbert space, and then acts as a unitary constrained QCA inside it.** That is the setting of PXP-type models, and it is not something the six rules do.

## 6.2 The induced unitary is PXP with the spin flip replaced by a Hadamard

§3 says the channel restricted to an attractor is *some* unitary. For these two it is an identifiable one.

**Proposition 7.** *On the largest attractor of $W_{73}$ and of $W_{109}$, at $N=9$ and 11, the induced one-period map equals, to machine precision ($\leq 1.7\times10^{-16}$ entrywise), the brickwork product over the even then the odd sublattice of the blockade-controlled Hadamard*

$$U_i \;=\; P^a_{i-1}\,H_i\,P^a_{i+1} \;+\; \big(\mathbb{1}-P^a_{i-1}P^a_{i+1}\big),$$

*with $a=0$ (fire only when both neighbours are* empty*) for $W_{73}$ and $a=1$ (both neighbours* occupied*) for $W_{109}$; sites beyond the chain count as 0. The map is unitary on that subspace to $\|AA^\intercal-\mathbb{1}\|\leq 4.4\times10^{-16}$.*

### Where that operator comes from: it is the coherent parent

The reference operator above was not pulled out of the air, though it was identified numerically before it was understood. Read the gate words in slot order (00), (01), (10), (11):

$$W_{201}=\texttt{VIII}\quad W_{73}=\texttt{VIID},\qquad W_{108}=\texttt{IIIV}\quad W_{109}=\texttt{EIIV}.$$

$W_{201}$ fires its Hadamard exactly when both neighbours are empty — it *is* the Floquet $P^0HP^0$ model — and $W_{108}$ fires when both are occupied. Each dissipative rule is its coherent parent with a single identity replaced by a reset. So the blockade projector in the proposition is not a coincidence: $a$ is inherited from whichever slot the parent's Hadamard occupies, and the induced unitary is literally the parent rule. (Observation due to the Tier-1 session, R24, which also gives the argument: $W_{73}$'s `D` sits in slot (11), so it acts only at a site whose neighbours are both 1 — on the no-adjacent-1s space such a site is itself 0, where `D` is the identity.)

Verified here independently on the *whole* constrained space, $N = 4, \ldots, 11$: no amplitude leaves it (escape exactly 0.0), the jump probability is 0 throughout, the one-cycle matrix is orthogonal to $4.4 \times 10^{-16}$, and it agrees with the parent's to $\leq 2.2 \times 10^{-16}$ in floating point.

**Invariance is not irreducibility, and the two rules differ.** The constrained space has dimension $F(N+2)$ for both, but they sit differently inside it:

| | $N = 9$ | $N = 11$ |
|---|---|---|
| $W_{73}$, no-`11` space | one enclosure, 89 | one enclosure, 233 |
| $W_{109}$, no-`00` space | four: $34, 21, 21, 13$ | four: $89, 55, 55, 34$ |

For $W_{73}$ the constrained space *is* the attractor, irreducibly. For $W_{109}$ it splits into exactly four enclosures — all Fibonacci, summing to $F(N+2)$ with *no* transient states left over and no enclosure straddling the boundary — which is why its largest attractor is $F(N)$ and not $F(N+2)$. The cause is not the reset firing at the open boundary: we measured $P(\text{jump}) = 0$ everywhere on that space, boundary included. It is that the parent $W_{108}$ is already reducible there under `obc0`.

This is the PXP structure exactly — a projector-dressed single-site operator on a hard-core constrained chain — with the spin flip $X$ replaced by the Hadamard $H$, and in Floquet (brickwork) rather than Hamiltonian form. The constraint is preserved for the obvious reason: flipping a site whose neighbours are both empty can never create an adjacent pair, so $P^0 H P^0$ maps the no-two-adjacent-1s space to itself, and $P^1 H P^1$ does the same for no-two-adjacent-0s.

The two rules are *not* related by a particle-hole conjugation, which is worth stating because the constraints look like mirror images. Conjugating by a global $X$ would send $H \mapsto XHX$, and $XHX$ is a different reflection: the $W_{109}$ map tested against a $P^1(XHX)P^1$ brickwork misses by 0.71, while the same test with $H$ is exact. Both rules carry the *same* gate and opposite blockade projectors, and the `obc0` boundary pins 0 rather than 1, which is why the dimensions come out at $F(N+2)$ and $F(N)$ rather than agreeing.

## 6.3 And they are entangled

The membership matrix across the middle cut has rank 2 — the blockade constraint is a bond-dimension-2 condition, so the attractor is *not* a product subspace, and states in it need not be product states. They are not: the half-chain entropy of the trajectory state deep in the attractor is

| | $N = 9$ | 11 | 13 | 15 |
|---|---|---|---|---|
| $W_{73}$ | $0.624 \pm 0.121$ | $0.737 \pm 0.158$ | $0.773 \pm 0.178$ | $0.928 \pm 0.215$ |
| $W_{109}$ | $0.386 \pm 0.084$ | $0.616 \pm 0.117$ | $0.737 \pm 0.152$ | $0.751 \pm 0.166$ |
| the six | 0 | 0 | 0 | 0 |

(24 trajectories each, averaged over an 8-period window, errors are standard errors of the mean; the six rules read $\leq 9.3 \times 10^{-17}$, i.e. machine zero, at every entry.) The contrast with that row of exact zeros is the result. The apparent *growth* with $N$ is **not** resolved at this sample size — the rise from $N = 9$ to $N = 15$ is $1.2\sigma$ for $W_{73}$ and $2.0\sigma$ for $W_{109}$ — so we deliberately fit no scaling law to it. The claim is that the entropy is nonzero and $O(1)$, not that it obeys a volume

law. A run with enough trajectories to separate a volume law from a logarithm is the obvious next measurement and is not attempted here.

Their occupation profile also has no period: $\max_t |n(t+p) - n(t)|$ stays $O(1)$ for every $p \leq 8$, against machine zero at $p = 2$ for the six.

### 6.4 This is the structural counterpart of the MPO result

The dissipative MPO sweep found an $N$-independent operator Schmidt rank for all six — $\chi_{\mathrm{sat}} = 53$ (28, 70), 18 (29, 71), 67 (199, 157) — and exponential growth for 73 and 109. The structure above is why. The asymptotic map of the six is a sum over a *regular* tiling language of *product* operators, and a regular language has a finite transfer matrix, so the bond dimension cannot grow with $N$. For 73 and 109 the asymptotic map carries an entangling unitary on a constrained space, and nothing bounds it. We state this as the mechanism, not as a derivation: the particular values 18, 53, 67 are not accounted for here.

## 7 How much is actually protected

Everything above counts *one* enclosure at a time. But §4.4 showed the whole recurrent space carries a single unitary, so the protected space is the span of *all* recurrent basis states, of dimension $D = \sum_k n_k$, and that is a strictly bigger object than the largest enclosure. Each rule therefore has three exponents, not one, and they are all different:

| | number of enclosures | largest enclosure | protected dimension $D$ |
|---|---|---|---|
| the six | $\rho = 1.3247$ | $2^{1/3} = 1.2599$ | **1.5214** |
| | $x^3 = x + 1$ | exact | $x^3 = x + 2$ |
| $W_{73}$, $W_{109}$ | $\psi = 1.4656$ | $\varphi = 1.6180$ | **1.8393** |
| | $x^3 = x^2 + 1$ | $x^2 = x + 1$ | $x^3 = x^2 + x + 1$ |

The right-hand column is measured, then identified as the dominant root of an exact integer recurrence fitted to $N = 4, \ldots, 13$ and verified on every term:

| | |
|---|---|
| $W_{73}$, $W_{109}$ | $D(N) = D(N-1) + D(N-2) + D(N-3)$ (tribonacci) |
| $W_{28}$, $W_{70}$ | $D(N) = D(N-1) + D(N-2) + D(N-3) - 2D(N-4)$ |
| $W_{157}$, $W_{199}$, $W_{29}$, $W_{71}$ | $D(N) = D(N-2) + 2D(N-3)$ |

with $W_{73}$ running $13, 24, 44, 81, 149, 274, 504, 927, 1705, 3136$ and $W_{109}$ running $9, 17, 31, 57, 105, 193, 355, 653, 12$
The last two recurrences differ but share a characteristic root, the real root of $x^3 = x + 2$.

In qubits per site the protected space holds $\log_2 1.8393 = 0.879$ for 73 and 109, against $\log_2 \varphi = 0.694$ inside the largest blockade sector alone, and $\log_2 1.5214 = 0.605$ for the six. The gap between 0.879 and 0.694 is exactly the cross-enclosure coherence of §4.4: a per-attractor accounting throws it away, and it is the larger part of what survives. (The tribonacci growth and the qubits-per-site framing are due to the Tier-1 session, R24; the two recurrences for the six and their shared root $x^3 = x + 2$ are measured here.)

None of this is protection in the coding sense — §4.4 already showed the distance is 1 — but it is the right size for the noiseless subsystem against the machine's own resets.

## 8 Answering the question as asked

1. **Is the stripe pattern real?** Yes, and it is a wall tiling. The stripes are the walls, the gaps are tiles of width 2 or 3, and which tiling you get is chosen by the noise from a set that grows like $\rho^N$.

2. **Is the oscillation real?** Yes, and it is exactly period 2: one dark qubit per wide tile running $H^t$, coherent and immune to the dissipation because the attractor is decoherence-free. It is a single-shot effect and averages away.

3. **How does it relate to the terminal-SCC digraph?** Directly. The terminal SCC is a complete digraph on a subcube, and that is the graph-theoretic shadow of "$k$ independent dark qubits": completeness = the bits are redrawn independently and uniformly; the subcube = they are independent *sites*; the frozen complement = the walls.

4. **Are the six trivial?** Asymptotically yes — product state, zero entanglement, memoryless, a decoupled period-2 flicker. What is not trivial is the *fragmentation*: an exponential number of decoherence-free subspaces surviving dissipation, selected by the measurement record. The interest is in the selection, not in the asymptotic dynamics.

5. **What happens in 29 and 71?** A weak component holds several enclosures. From a shared state the fate is settled in one period by $k$ independent fair coins, uniform on $2^k$ outcomes, $k \in \{1, 2\}$. The coin picks the width of one tile and requires both an `E` and a `D`. Not a bifurcation: no parameter, ratio pinned at 1/2, same attractor set as 157 and 199.

6. **Are 73 and 109 more interesting?** Decisively. Their attractors are the hard-core constrained spaces of dimension $F(N + 2)$ and $F(N)$ — not product subspaces, rank 2 across a cut, hosting an entangled and aperiodic constrained unitary. The induced unitary is identified exactly: blockade-controlled Hadamards, PXP with $X \to H$ in Floquet form. Dissipative preparation of an exponentially large decoherence-free subspace carrying a PXP-type constrained dynamics is worth a paper; a set of decoupled flickering qubits is not.

# 9 Reproduction

```
python -m qca_fragmentation.scaling.stripes --rebuild
python -m pytest qca_fragmentation/tests/test_stripes.py -q
```

All numbers come from `analytics/stripes_obc0.json`. The exact statements (subcube, tiling grammar, absorption law, decoherence-free test) are full enumerations of $2^N$; the entanglement and period-2 numbers are state-vector quantum trajectories with the per-site two-Kraus split of `core.cycle._split_site`, sampled.